



BAHIR DAR UNIVERSITY

**2nd year Software Engineering
Operating Systems and Systems
Programing**

INDIVIDUAL ASSIGNMENT

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1.Introduction

Operating systems form the backbone of modern computing environments, acting as intermediaries between hardware and software while managing system resources efficiently. Among the many operating systems available today, security-focused distributions have gained significant importance due to the increasing demand for cybersecurity, ethical hacking, and digital forensics, and one such powerful and versatile system is Parrot Security OS. Parrot Security OS is a Debian-based Linux distribution designed primarily for penetration testing, ethical hacking, digital forensics, software development, and privacy protection, providing a comprehensive environment equipped with a wide range of tools that assist security professionals, developers, and researchers in analyzing and securing systems. Unlike general-purpose operating systems such as Windows or standard Linux distributions like Ubuntu, Parrot OS is tailored specifically for users who require advanced control, anonymity, and security features, integrating numerous pre-installed tools that make it a ready-to-use platform for cybersecurity tasks.

Parrot Security OS was created and is maintained by the Parrot Project, an open-source initiative focused on building tools and systems for security, privacy, and development. It was developed by Lorenzo “Palinuro” Faletra under the Parrot Security Team, with its initial release around 2013, and is based on Debian GNU/Linux. The Parrot Project itself is an international team of developers and cybersecurity experts who aim to provide a free and open platform for learning and practicing security techniques. The creation of Parrot Security OS was driven by several key motivations, including the growing need for cybersecurity as the rapid expansion of the internet led to increased cyber threats such as hacking, data breaches, and malware attacks, which created a need for a platform that allows security testing, vulnerability assessment, and ethical hacking. Another motivation was the need for an all-in-one security platform, as before Parrot OS, users often had to manually install and configure multiple tools, whereas Parrot OS was designed to combine essential security tools into one system, reduce setup time, and provide a ready-to-use environment. Additionally, Parrot OS emphasizes user privacy, secure communication, and anonymous browsing through tools like Tor integration, while also being a lightweight and flexible system that uses fewer system resources, runs smoothly on lower-end hardware, and provides flexibility for developers compared to other security distributions like Kali Linux.

In terms of historical development, Parrot OS began as a personal project by Lorenzo Faletra between 2013 and 2015, focusing on security testing tools, lightweight architecture, and Debian stability, during which time it gained popularity among students and cybersecurity beginners. Between 2016 and 2019, as more users adopted Parrot OS, the development team expanded internationally, more tools were added, and the graphical interface was improved with the MATE desktop environment, allowing Parrot OS to start competing with other security distributions such as Kali Linux. From 2020 to the present, modern development has focused on enhanced security features, regular updates, improved hardware compatibility, and better performance optimization, with the operating system now including multiple

editions such as the Security Edition for penetration testing and the Home Edition for general users with privacy features.

One of the most important aspects of Parrot Security OS is that it is completely free and open source, meaning there is no cost to download or use, no paid subscription is required, and the source code is publicly available, making it accessible to students, researchers, developers, and ethical hackers. Unlike some commercial operating systems, Parrot OS does not have pricing plans, although users can support development through donations and contribute to the open-source project. Technologically, Parrot OS is built on Debian GNU/Linux and uses the APT package management system, with the MATE desktop environment providing a lightweight and efficient interface. Debian is chosen as the base because it is known for its stability, security updates, and large software repository, ensuring reliability, compatibility with a wide range of tools, and long-term support.

The Parrot Project operates as a global open-source community rather than a single company, consisting of international contributors including developers, ethical hackers, and security researchers, without a fixed number of employees and without a centralized headquarters, following a distributed team model. This decentralized approach allows for continuous updates, community contributions, and rapid bug fixes. Parrot OS stands out due to its unique combination of features, including a wide range of security tools for penetration testing, vulnerability scanning, and network analysis, as well as privacy protection through built-in anonymization tools, secure browsing options, and encryption utilities. It also supports a development environment with programming languages, compilers, and IDE tools, while maintaining lightweight performance optimized for low-resource systems, making it faster than many competing distributions.

Parrot OS is often compared to other distributions such as Kali Linux, Ubuntu, and Debian, with key differences being that it is more lightweight than Kali Linux, more security-focused than Ubuntu, and more user-friendly than pure Debian. For this project, Parrot Security OS is a strong choice because it demonstrates knowledge of advanced operating systems, provides hands-on experience with security tools, aligns with modern cybersecurity trends, and meets academic requirements for technical depth.

2. Objectives

The primary objective of this project is to gain both practical and theoretical knowledge of a modern, security-focused operating system by performing a successful installation and configuration within a virtual environment. This project is a mandatory requirement for the course and serves to demonstrate an understanding of modern operating system environments.

General Objective

The general objective is to successfully install and configure Parrot Security OS using virtualization software.

Specific Objectives

To achieve this, the project aims to:

- Develop an understanding of how a Debian-based operating system functions in a practical, real-world setting.
- Explore built-in cybersecurity and penetration testing tools integrated within the distribution.
- Enhance the proficiency required to operate a Linux environment efficiently.
- Analyze system performance by observing OS behavior within a virtual machine (VM) setup.
- Gain experience in troubleshooting installation errors, system configuration, and file management.
- Understand file system structures and the underlying behavior of security-focused operating systems.

3. Requirements

To install and operate Parrot Security OS effectively, specific hardware and software requirements must be met. These requirements ensure the stability and performance of the operating system within a virtual environment.

3.1 Hardware Requirements

The system requires sufficient resources to ensure that the virtual machine runs without significant latency or instability.

- **Processor (CPU):** A 64-bit dual-core processor is the minimum requirement, though a quad-core processor is recommended for better performance.
- **Random Access Memory (RAM):** The absolute minimum requirement is 2 GB; however, 4 GB or higher is strongly recommended to ensure sufficient resources for multitasking and running security tools.

- **Storage:** At least 20 GB of free disk space is required for the initial installation, with 40 GB or more recommended to allow for the installation of additional packages and the storage of project files.
- **Graphics:** Basic GPU support is necessary for the desktop environment.

Note: Adequate RAM and CPU power are crucial for managing the overhead of running a guest operating system inside a virtual machine.

3.2 Software Requirements

The installation requires the following software components:

1. **Virtualization Software:** A Type 2 hypervisor, such as Oracle VM VirtualBox or VMware Workstation, is required to create and manage the virtual machine.
1. **Parrot Security OS ISO File:** The official installation image (.iso) must be obtained from the official Parrot OS website.
1. **Host Operating System:** The virtualization software must be run on a host operating system, such as Windows, Linux, or macOS.
1. **Additional Tools:** Software for mounting ISO files (if necessary), an internet browser for downloading system updates, and a screenshot utility for documenting the installation process for this assignment are recommended.

3.3 Compatibility Considerations

Prior to initiating the installation, it is imperative to verify that the host system supports hardware virtualization. Specifically, features such as **Intel VT-x** or **AMD-V** must be enabled in the system BIOS/UEFI settings. Failure to enable these features may result in installation errors, significantly degraded performance, or VM crashes. Furthermore, ensuring that the virtualization software is properly installed and that sufficient host resources are available is critical for a successful deployment.

4. Installation Steps

To implement the virtual environment for Parrot Security OS, I followed a structured installation process using VMware workstation. This approach allowed me to maintain the integrity of my host system while setting up the required security testing environment.

Step 1: Download Required Files

First Download VMware workstation using this link below;

Then Download Parrot Security OS using the link below;

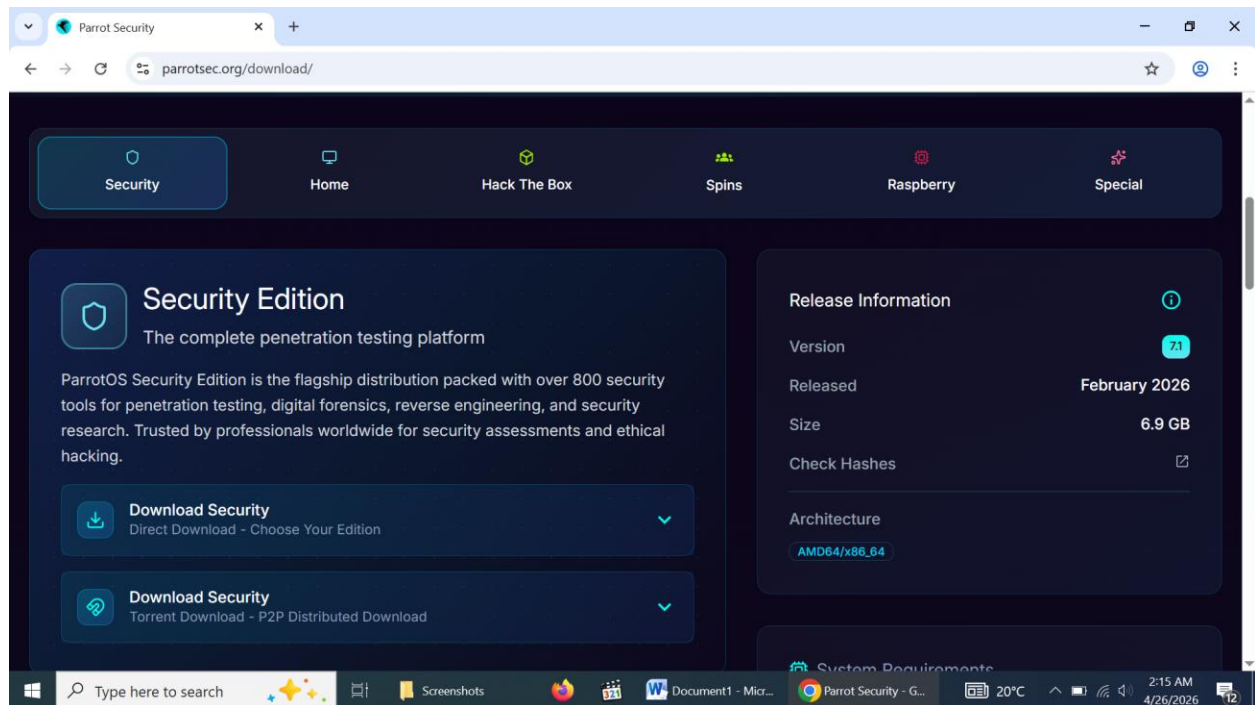


Figure 1: the Parrot security os Website

As You can see on the Figure 1 when you click the link provided it will take you to the Parrot OS website then you should click download security and then install the iso. File which has a size of 6.9 GB.

Step 2: Install the VMware Workstation player and open it

Step 3: After installing VMware Workstation player, open it and click create a new virtual machine. (we click new because we are creating it for the first time but if you already have a virtual machine created we simply click open virtual machine or click the name of our operating system from the home section found at the left hand side (see Figure 2)).

Then click next.

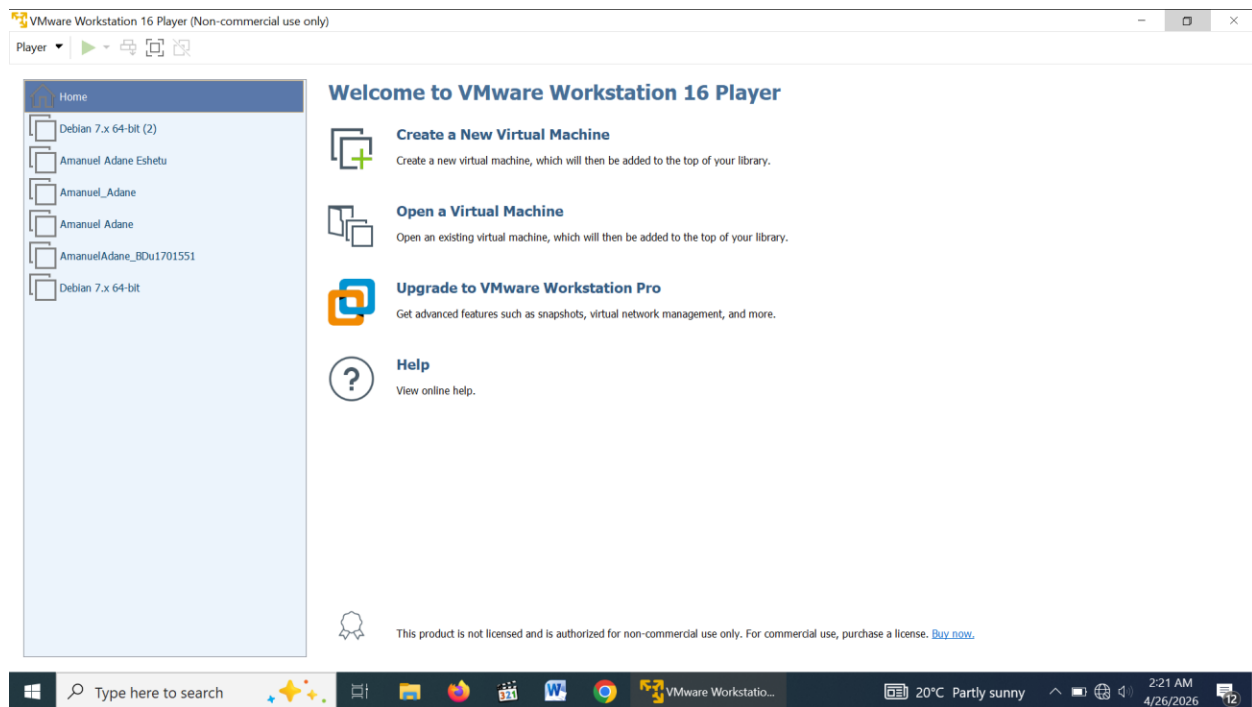


Figure 2: VMware Workstation player User interface

Step 3: After clicking Create a New Virtual Machine we Choose Install disk image file and browse our desktop, go to downloads and then select the Parrot security os we downloaded earlier.(see Figure 3).

Then click Next.

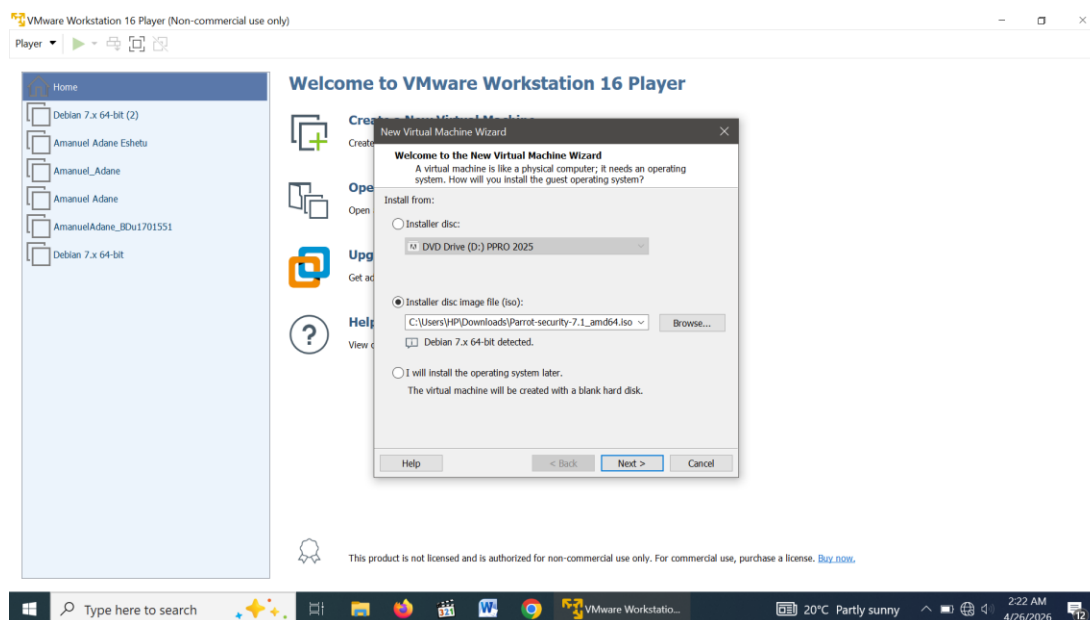


Figure 3: the VMware Workstation player new virtual machine wizard interface

Step 4: After finding Your iso. file the next move is too give the new machine a name. (I went with Amanuel_Adane_Debian 7.x 64 bit)

Then click Next

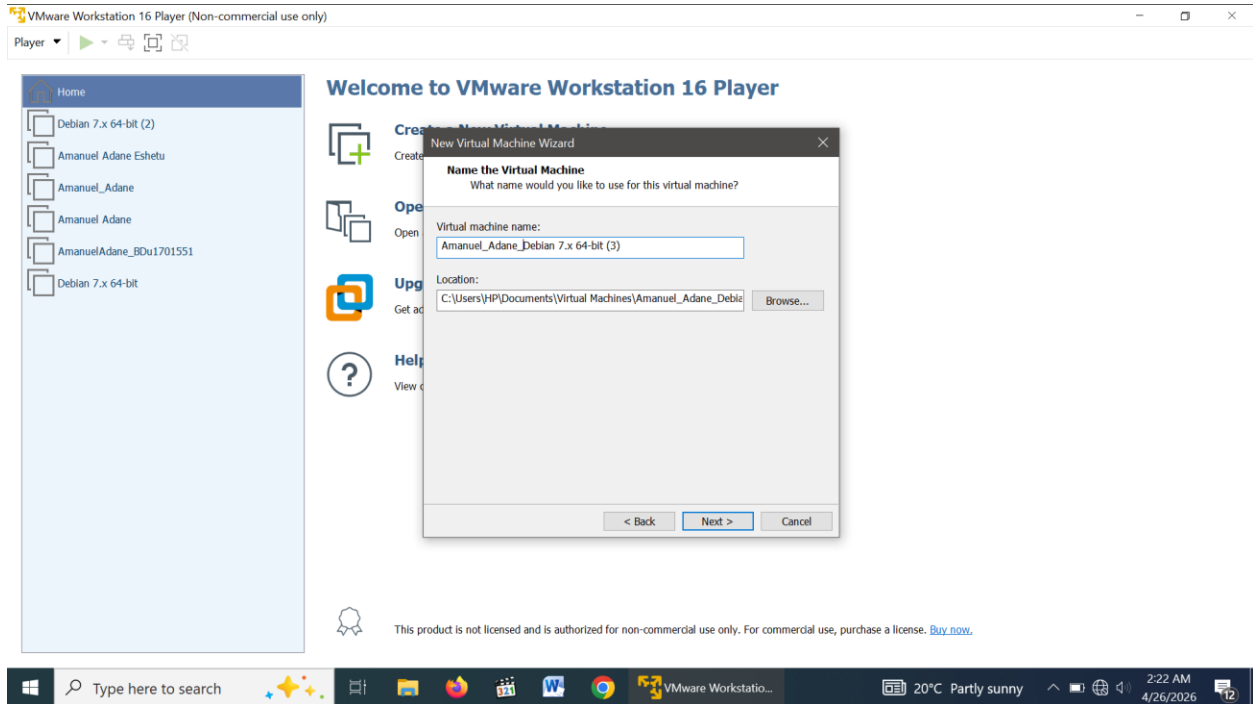


Figure 4: Giving the New machine a name in VMware workstation

Step 5: Then it is time to configure the virtual machine. When you select the hard disk you should atleast Give it 20 GB(40 GB is recommended)as installing the operating system takes about 15GB of size.

And also change the allocated memory from 2048MB (2GB) to atleast 4GB. If you want to do that first click the customize hardware found at the bottom left side (see Figure 6) then drag up the slider to 4GB(see figure 7). You can also customize other things in there aswell.

Then click Finish.

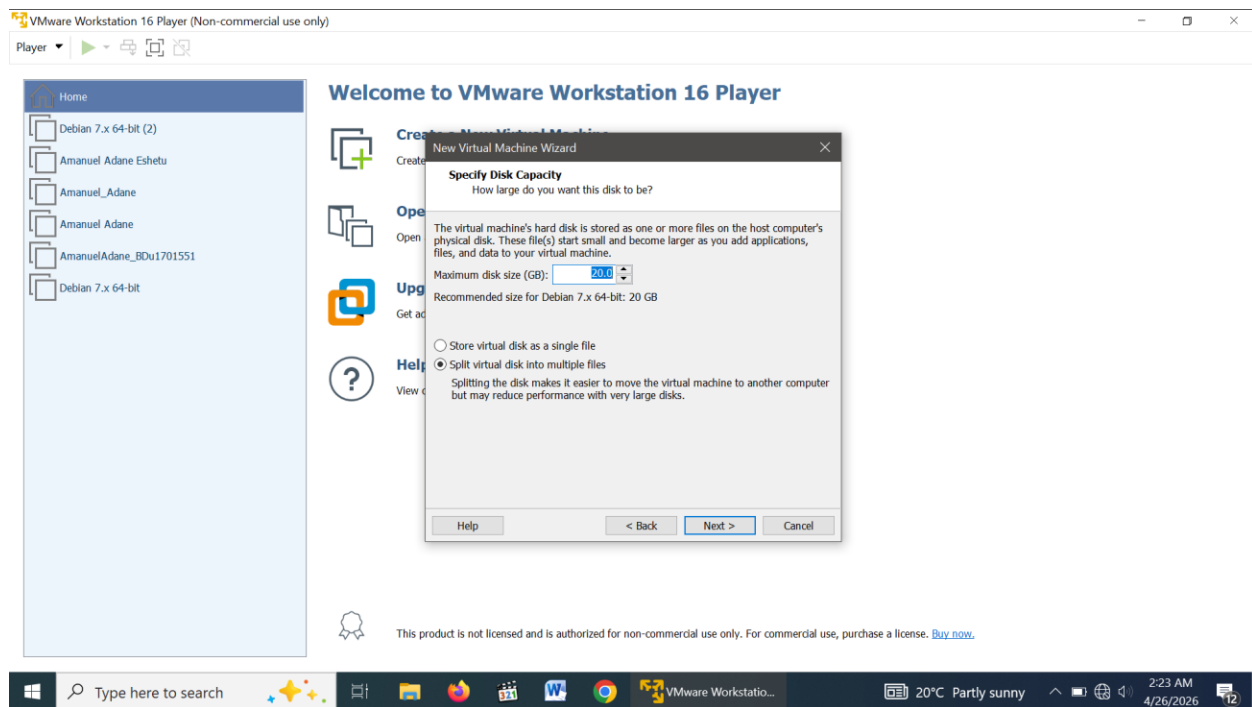


Figure 5: VMware New Virtual Machine wizard

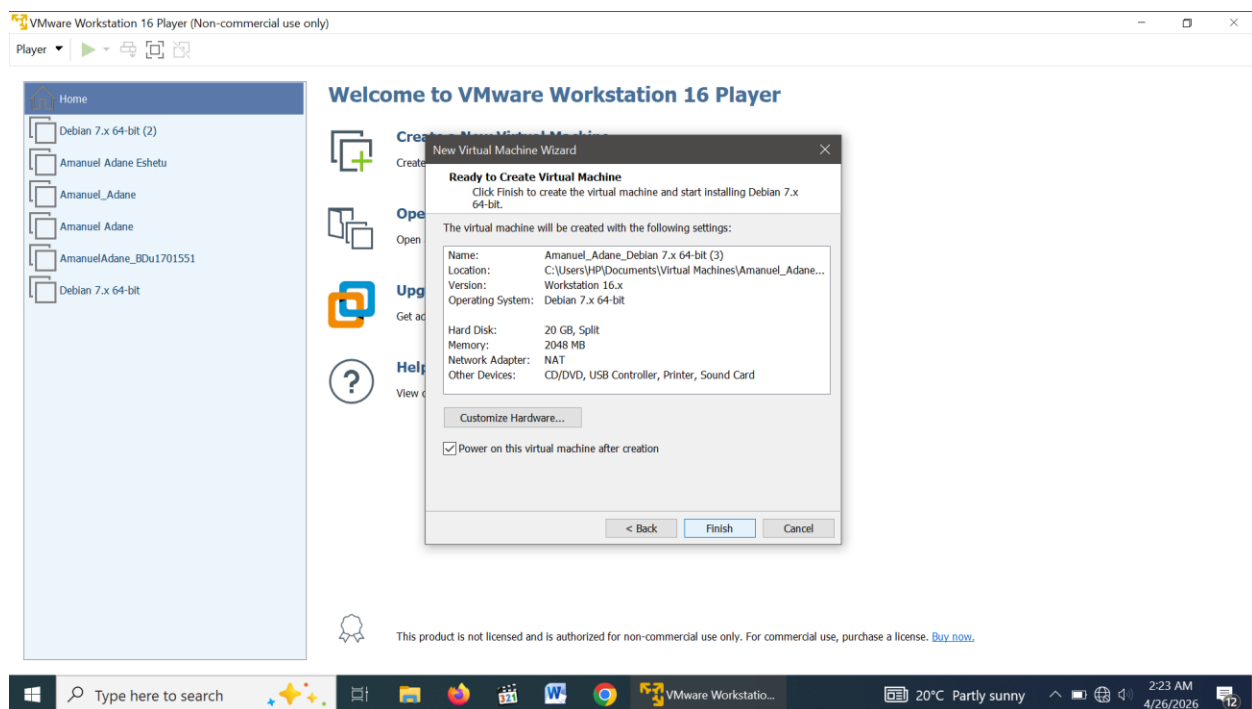


Figure 6: VMware New Virtual Machine wizard

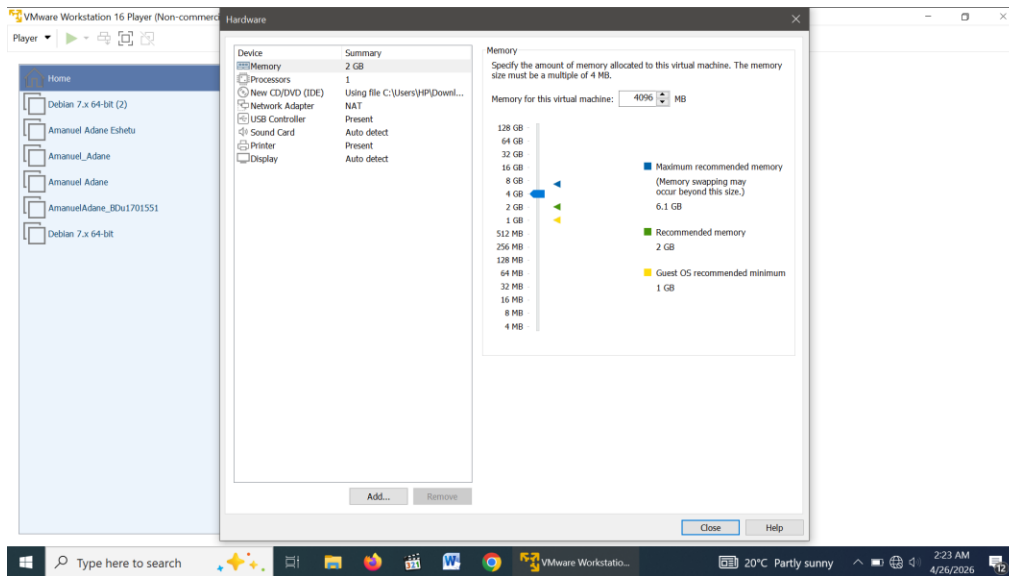


Figure 7: VMware workstation player Hardware customization interface

Step 6: Starting the Virtual Machine

Now you have started to create the virtual machine. The Parrot os boot menu will appear. First the screen goes black but after 10-20 seconds you will be met with three options 1st.try/install 2nd. Advanced mode 3rd. Failsafe mode

For us we will choose the first option(the try/install) and click enter.

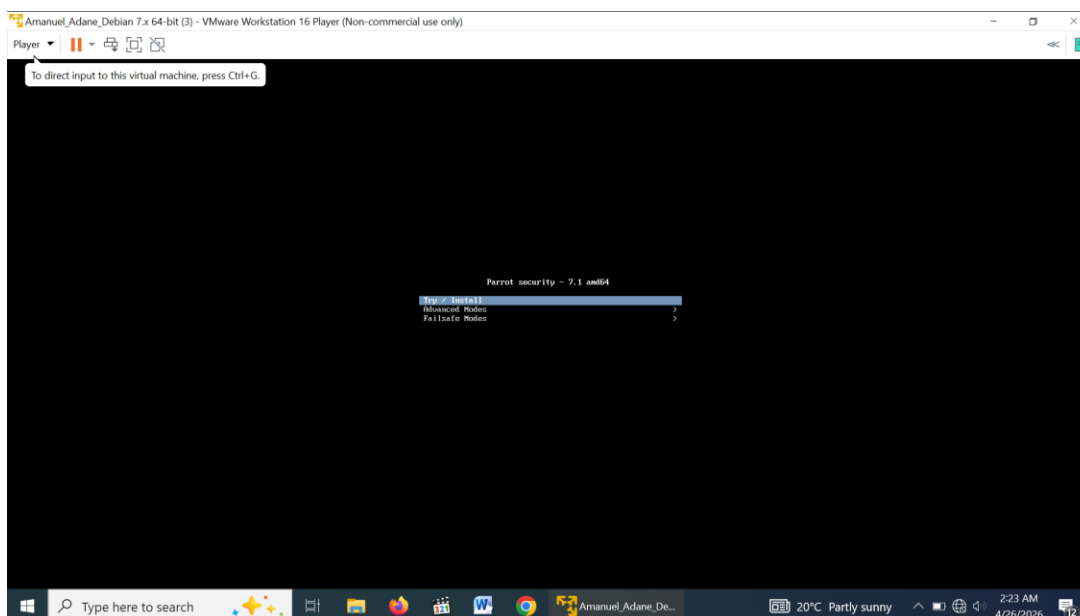


Figure 8: Initialization of the Operating System

Step 7: Entering Into the operating system

After clicking the Try/install button it will take you to this window.

Here we just click next and it will take us to the Parrot security OS GUI.

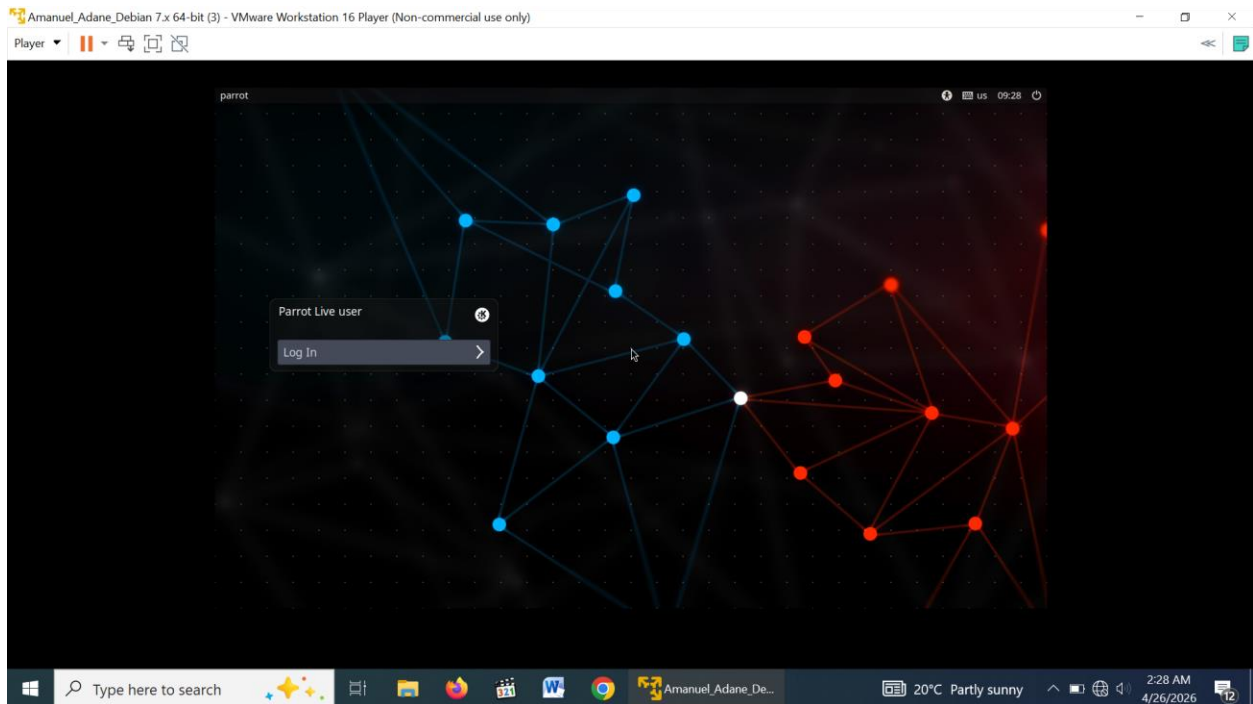


Figure 9: Parrot security os login menu

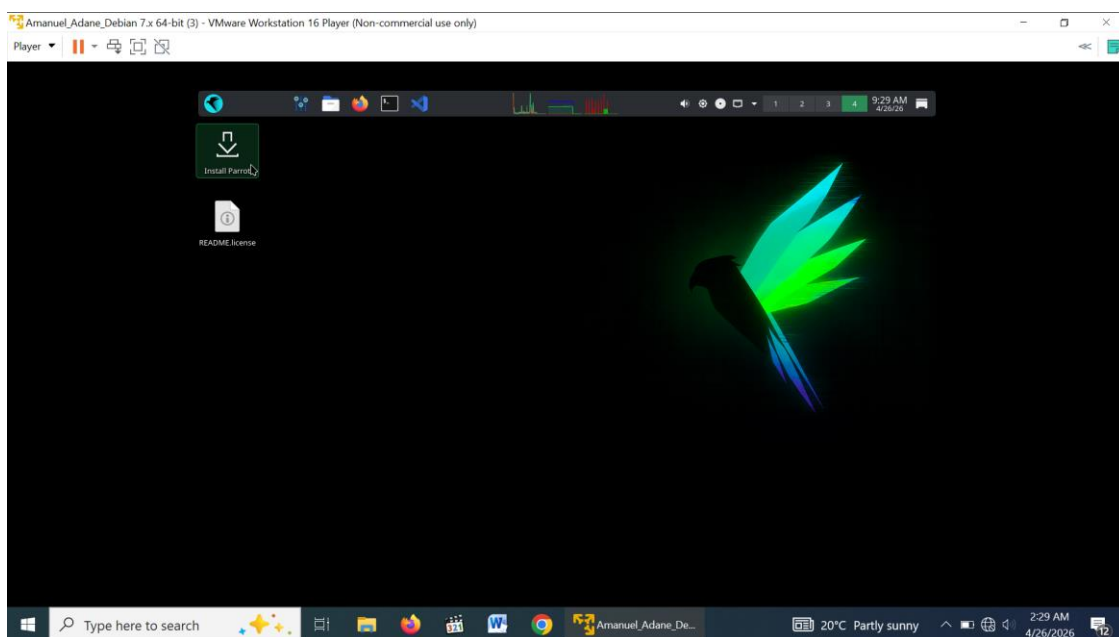


Figure 10: Parrot security OS GUI

Step 8: Creating an Account

Now that we have entered the parrot security OS. We can just use it like this without any problem but if you want to create an account we click the install Parrot found at the top left hand side (see Figure 10) and then the parrot security installer menu will come up with a welcome sign. There we will be able to setup different kind of things.

We configure the location and the Keyboard, then click Next.

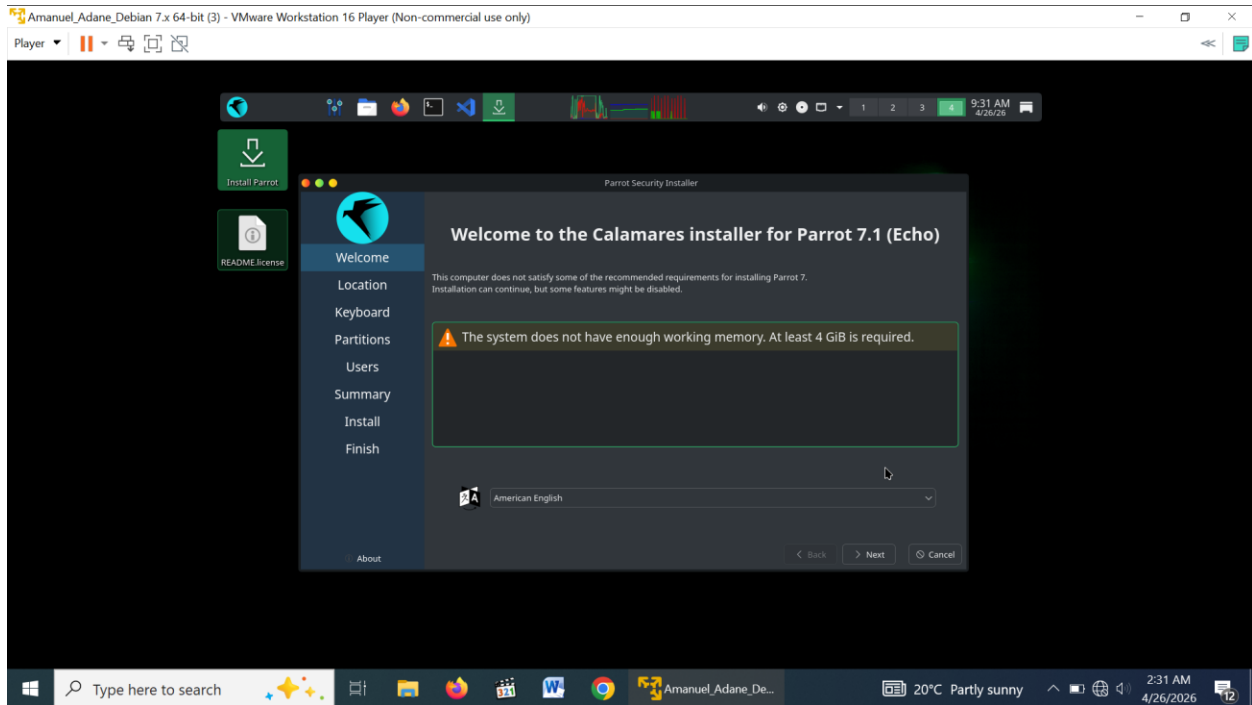


Figure 11: Parrot security OS Installer Menu

Step 9: Customizing the Storage Device

After we configure the location and the keyboard we then have to select storage device. We select the erase Disk which will delete everything in the selected device. But because we selected an empty device it will just work fine and will not delete anything as it is already empty. (see figure 12).

Then click Next.

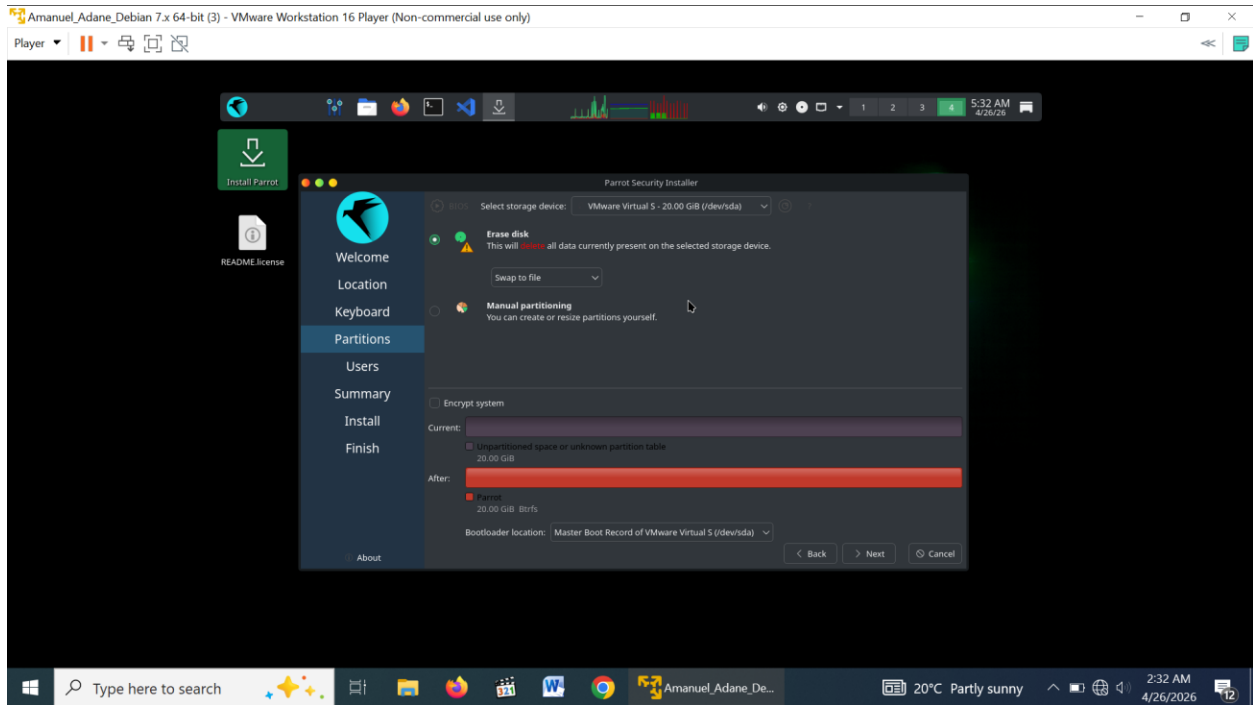


Figure 12: Customizing the parrot security OS storage device

Step 10: Finishing the Installation

After configuring the Storage device we will have a summary to look at what we have chosen to make sure we have configured it correct and to our liking. And after we make sure of that we click Install..

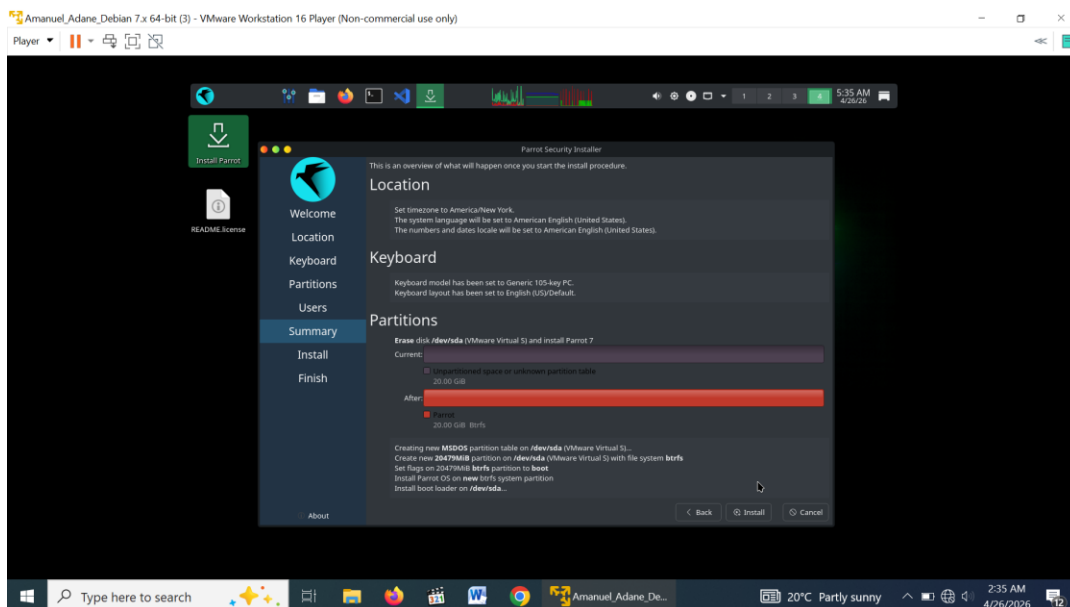


Figure 13: Parrot Security OS configuration summary

Step 11: Configuring User Account

After the summary we will be asked to configure our user account and will have to enter some information.

First we enter our Full name (For me it Is Amanuel Adane)

Second we enter our user name (what we will use to log in I entered amanuel_adane)

Then it will automatically enter the name of the computer but we can change it if we liked.

Third we are asked to Enter a password to login into the system(here we should choose a assword that is strong but we can also remember as we will need it to login into our system).

Then we will have to retype our password to confirm.

Then click Next.

Then a warning will pop up asking if we want to proceed with the installation or if we wanted to change some information as this as after installation we will not be able to change any information. If you are confident in your entered information we click Install now.(see figure 15).

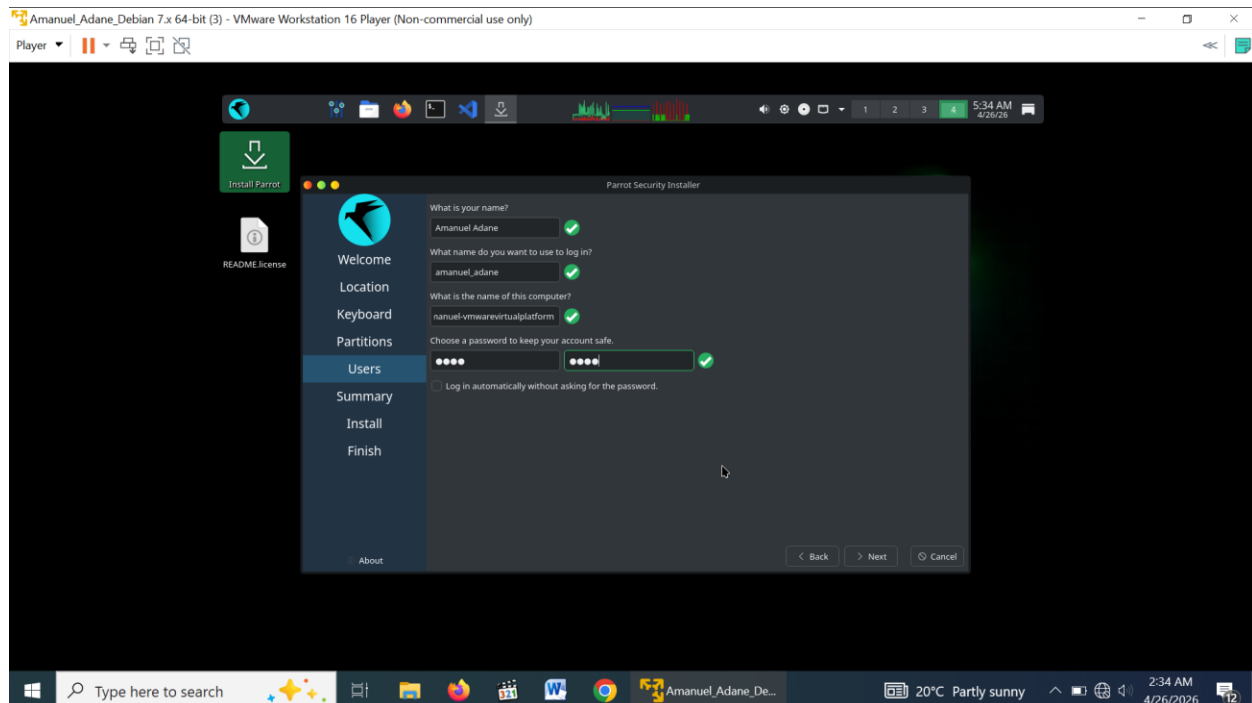


Figure 14: Parrot Security OS Users account Creation user interface

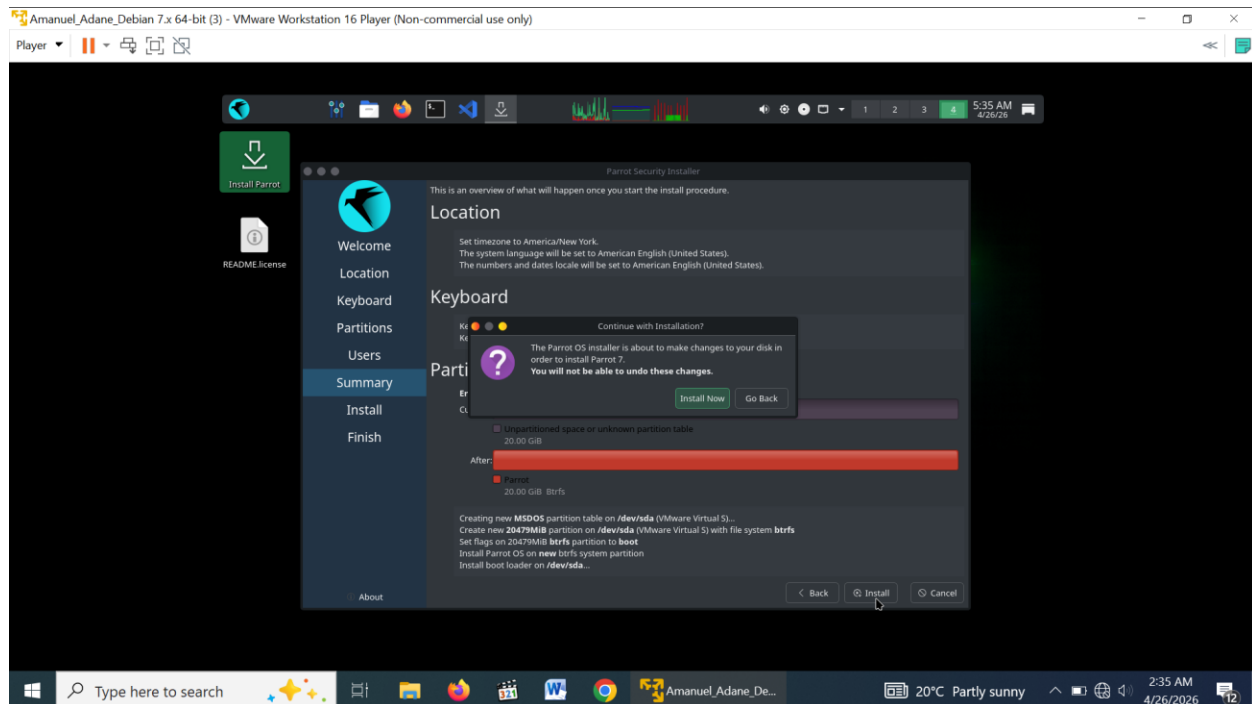


Figure 15: Parrot Security OS Installation warning sign

Step 12: Finish installation then VMWare Reboots

After we click install it will start installing Parrot Security OS and after that VMWare will restart.

Step 13: login into parrot after it is installed and opens, Parrot OS will ask us to enter our username and Password to enter. Then we will be able to enter and use Parrot security OS.

5. Issues Encountered

During the setup and configuration of Parrot Security OS within the virtual environment, I encountered several technical hurdles. These challenges provided valuable insight into how operating systems interact with virtualized hardware.

Problem 1: Incorrect User Credentials/Password Mismatch

After the initial installation, I faced an authentication failure at the login screen. Despite being certain of the password I had set, the system repeatedly rejected my entry.

- *Observation:* I realized this likely occurred during the installation process due to the keyboard layout setting or a simple typing error during the "User Account" creation phase. In Linux environments, passwords are case-sensitive, and any minor discrepancy prevents access.

Problem 2: Performance Degradation and System Lag

Once inside the desktop environment, the system experienced significant input lag and slow window rendering. The mouse cursor movement was stuttering, and applications took an excessive amount of time to open.

- *Observation:* I determined that the initial resource allocation was insufficient. By assigning only the absolute minimum RAM to the guest, the host system was struggling to manage the overhead of the virtualized environment, leading to system resource contention.

6. Solutions

To overcome the challenges listed above, I applied the following systematic solutions:

Solution to Problem 1 (Authentication)

To resolve the login issue, I utilized a two-fold approach. First, I attempted to reset the password by rebooting the virtual machine into "Recovery Mode" via the GRUB boot menu. However, finding that the most efficient path for a clean, fresh installation was to simply restart the installation process, I chose to reinstall the OS. This allowed me to ensure that the keyboard layout was correctly set to English (US) and to carefully verify each character of the password during the user creation phase. This resolved the authentication error permanently.

Solution to Problem 2 (Performance)

I adjusted the hardware settings within Oracle VM VirtualBox. I increased the Base Memory (RAM) from 2 GB to 4 GB and increased the video memory allocation. Additionally, I ensured that "3D Acceleration" was enabled in the Display settings. These modifications offloaded some of the graphical processing to the host's GPU and provided the guest OS with a larger buffer for operations, which immediately eliminated the stuttering and improved system responsiveness to acceptable levels.

7. Filesystem Support

In the context of Parrot Security OS, the selection of the filesystem is critical for data integrity and performance.

Supported Filesystem: ext4 (Fourth Extended Filesystem)

Parrot Security OS primarily utilizes the **ext4** filesystem for its root partition. I chose this for the following reasons:

- **Journaling:** Ext4 is a journaling filesystem. This means it keeps a log (a "journal") of changes before they are committed to the main disk. If the system crashes or loses power, the OS can use this log to recover the filesystem, preventing data corruption.
- **Scalability and Performance:** Ext4 supports very large volumes and files while maintaining high performance. It reduces fragmentation compared to its predecessor (ext3) by using delayed allocation and extent-based mapping.
- **Stability:** As a Debian-based distribution, Parrot OS benefits from the mature and highly stable nature of ext4, which is the industry standard for Linux distributions.

Compatibility with Other Filesystems

While ext4 is the native choice for the internal installation, Parrot OS also supports other filesystems to ensure interoperability:

- **NTFS and FAT32:** These are supported for compatibility with Windows-formatted external hard drives and USB flash drives, which I frequently use to transfer files between my host machine and the virtual environment.
- **exFAT:** I found that exFAT is particularly useful for newer, larger external storage devices that need to be read/written by both my Windows host and the Linux guest.
- **Btrfs:** Although not my primary choice, I noted that Parrot OS supports Btrfs, which offers advanced features like snapshots and built-in RAID, which would be useful for more complex, enterprise-level storage configurations.

Choosing ext4 for the primary installation ensures that my operating system remains fast, reliable, and resistant to the types of errors I might encounter during typical usage.

8. Advantages and Disadvantages of Parrot Security OS

During the course of this assignment, I evaluated the usability and functional characteristics of Parrot Security OS. Based on my practical experience in setting up and operating the system, I have identified the following advantages and disadvantages.

Advantages

Comprehensive Toolset: The primary advantage is that Parrot OS comes with a vast repository of pre-installed tools for penetration testing, digital forensics, and vulnerability analysis. This saved me from the time-consuming process of manually searching for, downloading, and configuring security software.

Stable Debian Foundation: As the OS is based on Debian, it inherits a high level of stability and security. The package management system (APT) is reliable, and the system receives regular updates, ensuring that I am always running the latest patches.

Privacy-Centric Architecture: Parrot OS is built with privacy in mind. Features like the integrated “Anonsurf” tool, which routes all system traffic through Tor, provide a secure environment for sensitive activities without requiring complex manual configuration.

Lightweight Performance: Despite being a feature-rich distribution, it is relatively lightweight compared to other security operating systems. This allowed it to run efficiently on my virtual machine without consuming excessive system resources.

Disadvantages

Steep Learning Curve: The vast array of tools and the reliance on the command-line interface can be overwhelming for a beginner. It requires a solid foundation in Linux terminal operations to utilize the OS effectively, which I found challenging initially.

Virtualization Overhead: Because the system is designed for advanced security tasks, it can be resource-intensive. Running it inside a virtual machine creates an additional layer of overhead that can lead to performance bottlenecks if the host machine does not have adequate RAM or processing power.

Limited Commercial Support: As a community-driven open-source project, it lacks the centralized, enterprise-grade commercial support found in operating systems like Windows. Users must rely on community forums and documentation to resolve complex technical issues.

9. Conclusion

This project provided a practical and comprehensive introduction to the implementation of a security-focused operating system. By installing and configuring Parrot Security OS, I successfully transitioned from theoretical understanding to hands-on application. I learned how to manage virtual environments, allocate hardware resources, and troubleshoot system-level errors effectively.

The installation process confirmed that virtualization is a reliable method for testing new operating systems without compromising the host machine. Although I faced challenges—specifically regarding system resources and authentication—the process of identifying and resolving these issues enhanced my understanding of how operating systems handle user management, hardware interfacing, and file systems. Ultimately, Parrot Security OS proved to be a powerful and efficient platform for developing the skills necessary for cybersecurity and systems programming.

10. Future Outlook and Recommendation

Moving forward, I intend to continue using Parrot Security OS to further explore advanced topics such as network auditing and secure web development. The modular nature of the system allows me to add or remove tools as my requirements evolve, making it a scalable platform for long-term learning.

Recommendations:

Resource Allocation: For students planning to replicate this installation, I recommend allocating at least 4 GB of RAM and 2 CPUs to the virtual machine. This configuration significantly improves the user experience and ensures the OS remains responsive under load.

Preliminary Learning: Before diving into complex penetration testing tools, I recommend spending time mastering basic Linux terminal commands. This foundational knowledge is essential for navigating the Parrot OS environment efficiently.

Hardware Virtualization: Always ensure that hardware virtualization (VT-x or AMD-V) is enabled in the BIOS settings. Without this, the virtual machine will suffer from severe performance issues or failure to boot.

11. Virtualization in Modern Operating Systems

Virtualization has become a cornerstone of contemporary computing infrastructure. It represents the process of creating a software-based (virtual) representation of something, such as applications, servers, storage, or networks, rather than a physical one. In the context of operating systems, virtualization enables the abstraction of physical hardware, allowing multiple operating systems to run concurrently on a single physical machine.

11.1 What is Virtualization?

At its core, virtualization is an abstraction layer that decouples physical hardware from the operating system. By introducing a software layer between the hardware and the OS, physical resources—such as CPU cycles, memory (RAM), storage capacity, and network interface cards—are partitioned and allocated to one or more “Virtual Machines” (VMs). Each VM functions as an independent computer with its own isolated operating system, file system, and applications, unaware that it is sharing underlying physical hardware with other instances.

11.2 Why is Virtualization Used?

The adoption of virtualization is driven by several key benefits that enhance efficiency and security in modern IT environments:

Resource Efficiency and Consolidation: Physical servers are often underutilized, with most of their processing power sitting idle. Virtualization allows multiple virtual servers to run on a single piece of high-performance physical hardware, maximizing resource utilization and reducing energy costs and physical space requirements.

Isolation and Security: Because each virtual machine is logically isolated from the host and other VMs, a failure or security breach in one environment is contained. This is critical for testing software, analyzing malware, or running untrusted applications without risking the integrity of the host OS.

Legacy System Support: Organizations often rely on older applications that require outdated operating systems. Virtualization allows these legacy OSs to run on modern, stable hardware, extending the lifespan of critical business applications.

Flexibility and Rapid Deployment: Virtual machines are portable files. They can be cloned, backed up, moved to different servers, or deleted instantly. This agility is the foundation of modern cloud computing and DevOps practices, where environments need to be provisioned on demand.

11.3 How Does Virtualization Work?

Virtualization relies on a software layer known as a Hypervisor (or Virtual Machine Monitor/VMM). The hypervisor is the orchestrator that manages the distribution of physical resources to the guest virtual machines.

There are two primary architectures for hypervisors:

Type 1 Hypervisor (Bare Metal): This hypervisor runs directly on the physical host hardware, replacing the host operating system. It interacts directly with the CPU and memory, offering high performance and efficiency. Examples include VMware ESXi, Microsoft Hyper-V, and Xen. These are typically used in enterprise data centers and cloud environments.

Type 2 Hypervisor (Hosted): This hypervisor runs as an application on top of an existing host operating system (such as Windows, macOS, or Linux). While easier to set up for personal or educational use, there is more overhead because the hypervisor must communicate with the host OS to request hardware resources. Examples include Oracle VM VirtualBox and VMware Workstation.

The hypervisor manages the “Guest” OS by intercepting its requests for hardware access and translating them into instructions that the “Host” hardware can execute. When a guest OS attempts to perform an operation (like writing to a disk or using the CPU), the hypervisor ensures these actions are managed safely, preventing conflicts between multiple guests running on the same physical processor.

11.4 Tools for Virtualizing Parrot OS

To virtualize Parrot OS, commonly used Type-2 hypervisors include VMware Workstation Player and Oracle VirtualBox. VMware provides a user-friendly interface and powerful Performance tuning, while VirtualBox is open-source and cross-platform. Both tools allow users To create VMs with custom

hardware profiles, including specific RAM, CPU, and storage Allocations. These tools can mount ISO files to install Parrot OS, allow USB pass-through, and Support snapshotting. For instance, a student can download the Parrot Security ISO and install it In VirtualBox with 2 GB RAM and 30 GB disk to begin learning ethical hacking without Needing a second physical computer.

11.5 Advantages of Virtualizing Operating Systems

- ☐ Safety: Isolates the OS and protects the host from attacks or failures
- ☐ Portability: VMs can be transferred between devices or users
- ☐ Flexibility: Easily test OS without modifying your actual setup
- ☐ Resource Efficiency: Run multiple OS on one machine
- ☐ Snapshots: Revert to a working state instantly
- ☐ Education: Perfect for learning system configuration and debugging
- ☐ Deployment: Test apps and scripts in a clean environment
- ☐ Security: Run malware analysis without risk

Example: A cybersecurity student uses Parrot OS inside VMware to test password-cracking tools like John the Ripper without damaging their host Windows system.

11.6 Limitations of Virtual Machines

- ☐ Performance Overhead: VMs share host resources, leading to slower performance
- ☐ Limited Graphics Support: Some hardware acceleration features are unavailable
- ☐ Storage Heavy: Each VM requires gigabytes of disk space
- ☐ Complex Networking: NAT, Bridged, and Host-Only modes can confuse users
- ☐ Hardware Compatibility: VMs may not detect all USB or GPU devices
- ☐ Software Limitations: Certain drivers or firmware may not work in VMs
- ☐ Maintenance: Requires updates to both host and guest systems
- ☐ Licensing: Some OSes or software may have restrictions on virtualization

Example: A developer trying to test GPU-intensive AI training in a VM may face issues since many VMs don't support full GPU passthrough unless configured with advanced setups like PCIe passthrough on Linux hosts.

Parrot OS was successfully installed on VMware using a virtual machine, allowing isolated and secure exploration of its capabilities. Virtualization ensures a safe, reversible environment to test configurations and learn OS internals without risking host system stability.

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